

**Q5 Motion under gravity A
level only**

Name: _____

Class: _____

Date: _____

Time: **493 minutes**

Marks: **408 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

In this question use $g = 9.81 \text{ m s}^{-2}$

A particle is projected with an initial speed u , at an angle of 35° above the horizontal.

It lands at a point 10 metres vertically below its starting position.

The particle takes 1.5 seconds to reach the highest point of its trajectory.

(a) Find u .

(3)

(b) Find the total time that the particle is in flight.

(3)

(Total 6 marks)

Q2.

In this question use $g = 9.81 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector $(40\mathbf{i} - 10\mathbf{j})$ metres.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that there are no resistance forces acting on the ball.

(a) Find the speed of the ball when it is at a height of 3 metres above its initial position.

(6)

(b) State the speed of the ball when it is at its maximum height.

(1)

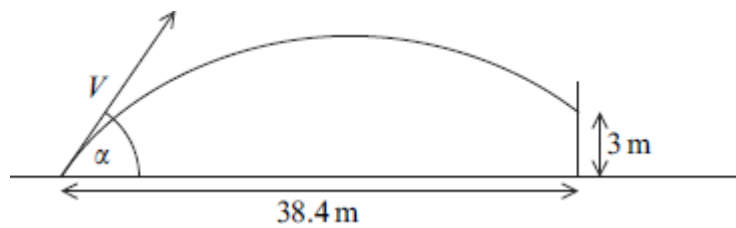
(c) Explain why the answer you found in part (b) may not be the actual speed of the ball when it is at its maximum height.

(1)

(Total 8 marks)

Q3.

A golf ball is hit from a point on a horizontal surface, so that it has an initial velocity $V \text{ m s}^{-1}$ at an angle α above the horizontal. The ball travels through the air and after 2.4 seconds hits a vertical wall at a height of 3 metres. The wall is at a horizontal distance of 38.4 metres from the point where the ball was hit. The path of the ball is shown in the diagram.

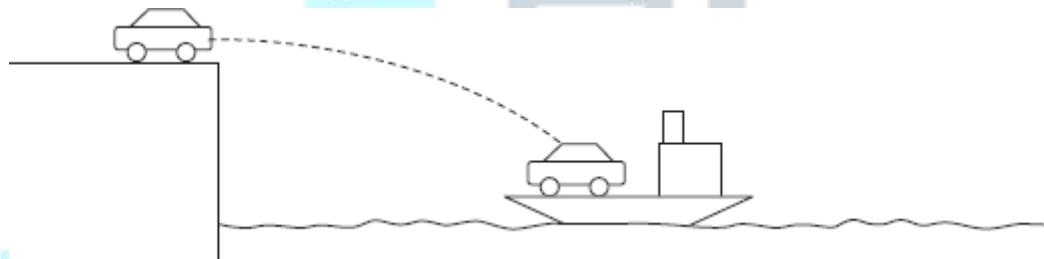


Assume that the weight of the ball is the only force that acts on it as it travels through the air.

- (a) Find the horizontal component of the velocity of the ball. (2)
 - (b) Find V . (5)
 - (c) Find α . (3)
- (Total 10 marks)**

Q4.

In a scene from an action movie, a car is driven off the edge of a cliff and lands on the deck of a boat in the sea, as shown in the diagram.



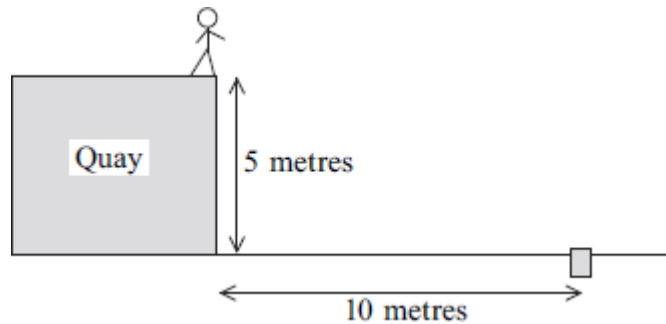
To land on the boat, the car must move 20 metres horizontally from the cliff. The level of the deck of the boat is 8 metres below the top of the cliff. Assume that the car is a particle which is travelling horizontally when it leaves the top of the cliff and that the car is not affected by air resistance as it moves.

- (a) Find the time that it takes for the car to reach the deck of the boat. (3)
 - (b) Find the speed at which the car is travelling when it leaves the top of the cliff. (3)
 - (c) Find the speed of the car when it hits the deck of the boat. (4)
- (Total 10 marks)**

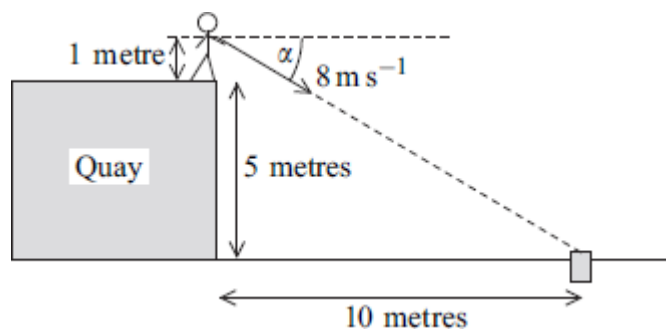
Q5.

A girl stands at the edge of a quay and sees a tin can floating in the water. The water level is 5 metres below the top of the quay and the can is at a horizontal distance of 10 metres

from the quay, as shown in the diagram.



The girl decides to throw a stone at the can. She throws the stone from a height of 1 metre above the top of the quay. The initial velocity of the stone is 8 m s^{-1} at an angle α below the horizontal, so that the initial velocity of the stone is directed at the can, as shown in the diagram.

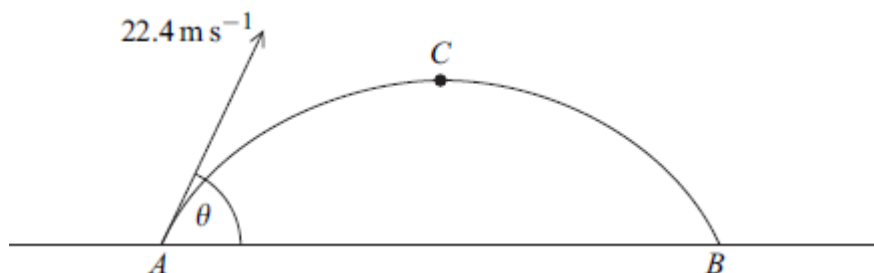


Assume that the stone is a particle and that it experiences no air resistance as it moves.

- (a) Find α . (2)
 - (b) Find the time that it takes for the stone to reach the level of the water. (6)
 - (c) Find the distance between the stone and the can when the stone hits the water. (4)
- (Total 12 marks)

Q6.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1} at an angle θ above the horizontal, as shown in the diagram.

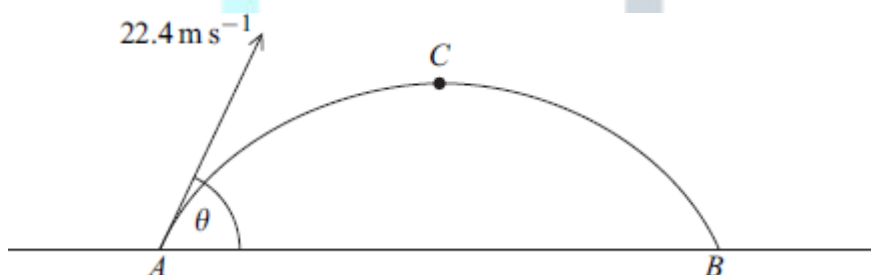


After 2 seconds, the particle reaches the point C , where it is at its maximum height above the surface.

- (a) Show that $\sin \theta = 0.875$. (3)
 - (b) Find the height of the point C above the horizontal surface. (3)
 - (c) The particle returns to the surface at the point B . Find the distance between A and B . (3)
 - (d) Find the length of time during which the height of the particle above the surface is greater than 5 metres. (5)
- (Total 14 marks)**

Q7.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1} at an angle θ above the horizontal, as shown in the diagram.



After 2 seconds, the particle reaches the point C , where it is at its maximum height above the surface.

- (a) Show that $\sin \theta = 0.875$. (3)
- (b) Find the height of the point C above the horizontal surface. (3)
- (c) The particle returns to the surface at the point B . Find the distance between A and B . (3)
- (d) Find the length of time during which the height of the particle above the surface is greater than 5 metres. (5)
- (e) Find the minimum speed of the particle. (2)

(Total 16 marks)

Q8.

(In this question, take $g = 10 \text{ m s}^{-2}$.)

A projectile is fired from a point O with speed u at an angle of elevation α above the horizontal so as to pass through a point P . The projectile travels in a vertical plane through O and P . The point P is at a horizontal distance $2k$ from O and at a vertical distance k above O .

- (a) Show that α satisfies the equation

$$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0 \quad (7)$$

- (b) Deduce that

$$u^4 - 20ku^2 - 400k^2 \geq 0 \quad (3)$$

(Total 10 marks)

Q9.

An arrow is fired from a point at a height of 1.5 metres above horizontal ground. It has an initial velocity of 12 m s^{-1} at an angle of 30° above the horizontal. The arrow hits a target at a height of 1 metre above horizontal ground. The path of the arrow is shown in the diagram.



Model the arrow as a particle.

- (a) Show that the time taken for the arrow to travel to the target is 1.30 seconds, correct to three significant figures. (5)
- (b) Find the horizontal distance between the point where the arrow is fired and the target. (2)
- (c) Find the speed of the arrow when it hits the target. (4)
- (d) Find the angle between the velocity of the arrow and the horizontal when the arrow

hits the target.

(2)

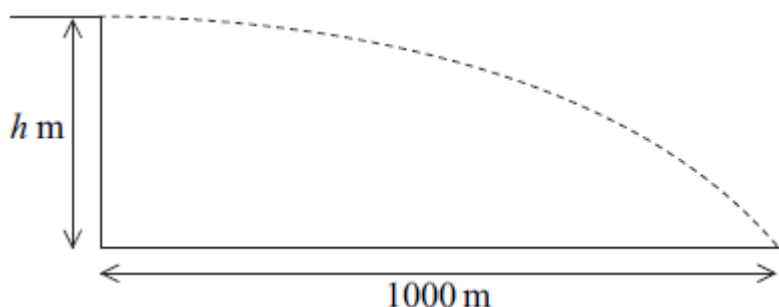
- (e) State one assumption that you have made about the forces acting on the arrow.

(1)

(Total 14 marks)

Q10.

A bullet is fired horizontally from the top of a vertical cliff, at a height of h metres above the sea. It hits the sea 4 seconds after being fired, at a distance of 1000 metres from the base of the cliff, as shown in the diagram.



- (a) Find the initial speed of the bullet.

(2)

- (b) Find h .

(2)

- (c) Find the speed of the bullet when it hits the sea.

(4)

- (d) Find the angle between the velocity of the bullet and the horizontal when it hits the sea.

(3)

(Total 11 marks)

Q11.

(In this question, use $g = 10 \text{ m s}^{-2}$.)

A golf ball is hit from a point O on a horizontal golf course with a velocity of 40 m s^{-1} at an angle of elevation θ . The golf ball travels in a vertical plane through O . During its flight, the horizontal and upward vertical distances of the golf ball from O are x and y metres respectively.

- (a) Show that the equation of the trajectory of the golf ball during its flight is given by

$$x^2 \tan^2 \theta - 320x \tan \theta + (x^2 + 320y) = 0$$

(6)

- (b) (i) The golf ball hits the top of a tree, which has a vertical height of 8 m and is at

a horizontal distance of 150 m from O .

Find the two possible values of θ .

(5)

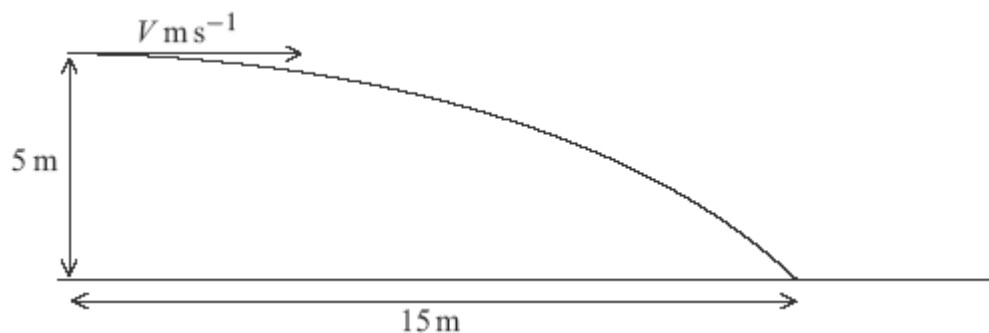
- (ii) Which value of θ gives the shortest possible time for the golf ball to travel from O to the top of the tree? Give a reason for your choice of θ .

(2)

(Total 13 marks)

Q12.

A ball is projected horizontally with speed $V \text{ m s}^{-1}$ at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.



- (a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures.

(3)

- (b) Find V .

(2)

- (c) Find the speed of the ball when it hits the ground.

(4)

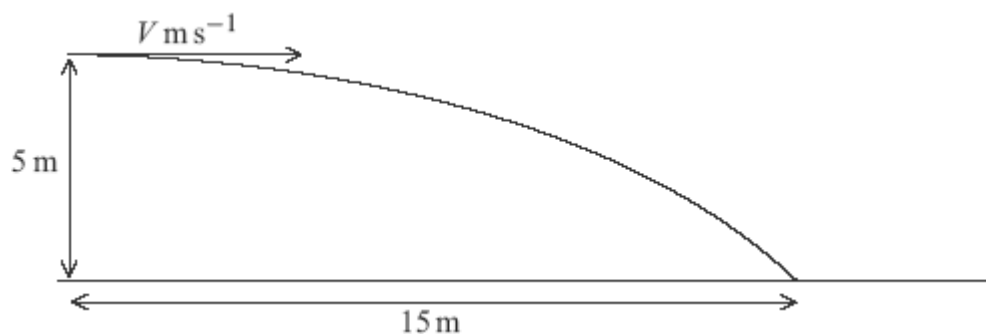
- (d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree.

(3)

(Total 12 marks)

Q13.

A ball is projected horizontally with speed $V \text{ m s}^{-1}$ at a height of 5 metres above horizontal ground. When the ball has travelled a horizontal distance of 15 metres, it hits the ground.

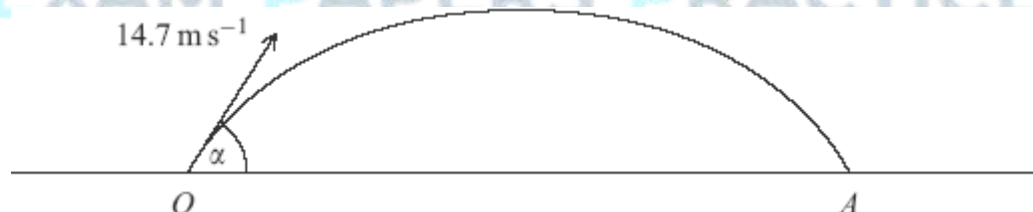


- (a) Show that the time it takes for the ball to travel to the point where it hits the ground is 1.01 seconds, correct to three significant figures. (3)
- (b) Find V . (2)
- (c) Find the speed of the ball when it hits the ground. (4)
- (d) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground. Give your answer to the nearest degree. (3)
- (e) State two assumptions that you have made about the ball while it is moving. (2)

(Total 14 marks)

Q14.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α above the horizontal. The path of the ball is shown in the diagram.

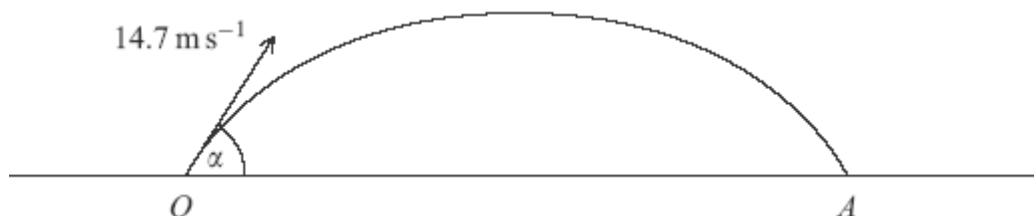


- (a) Show that the ball takes $\frac{3 \sin \alpha}{2}$ seconds to reach its maximum height. (3)
- (b) The ball reaches a maximum height of 7 metres.
 - (i) Find α . (5)
 - (ii) Find the range, OA . (3)

(Total 11 marks)

Q15.

A ball is struck so that it leaves a horizontal surface travelling at 14.7 m s^{-1} at an angle α above the horizontal. The path of the ball is shown in the diagram.



- (a) Show that the ball takes $\frac{3 \sin \alpha}{2}$ seconds to reach its maximum height. (3)
- (b) The ball reaches a maximum height of 7 metres.
- (i) Find α . (5)
- (ii) Find the range, OA . (3)
- (c) State two assumptions that you needed to make in order to answer the earlier parts of this question. (2)

(Total 13 marks)

Q16.

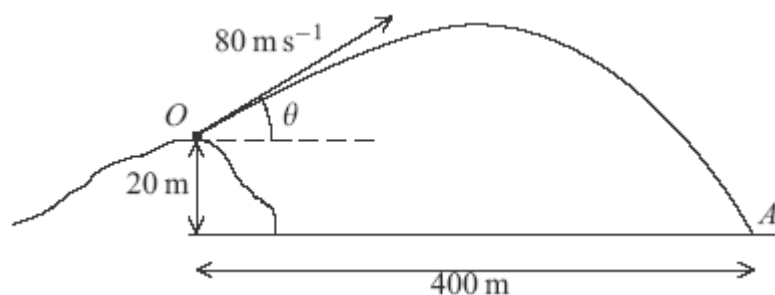
A projectile is fired from a point O on top of a hill with initial velocity 80 m s^{-1} at an angle θ above the horizontal and moves in a vertical plane. The horizontal and upward vertical distances of the projectile from O are x metres and y metres respectively.

- (a) (i) Show that, during the flight, the equation of the trajectory of the projectile is given by

$$y = x \tan \theta - \frac{gx^2}{12\,800} (1 + \tan^2 \theta)$$

(5)

- (ii) The projectile hits a target A , which is 20 m vertically below O and 400 m horizontally from O .



Show that

$$49 \tan^2 \theta - 160 \tan \theta + 41 = 0$$

(2)

- (b) (i) Find the two possible values of θ . Give your answers to the nearest 0.1° .

(3)

- (ii) Hence find the shortest possible time of the flight of the projectile from O to A .

(2)

- (c) State a necessary modelling assumption for answering part (a) (i).

(1)

(Total 13 marks)

Q17.

A football is kicked from ground level with an initial velocity of 30 m s^{-1} at an angle of 35° above the horizontal.

- (a) Find the maximum height of the ball above ground level.

(4)

- (b) Show that when the speed of the ball is 28 m s^{-1} , the magnitude of the vertical component of its velocity is 13.4 m s^{-1} , correct to three significant figures.

(4)

- (c) Find the angle between the velocity of the ball and the horizontal when the speed of the ball is 28 m s^{-1} .

(2)

(Total 10 marks)

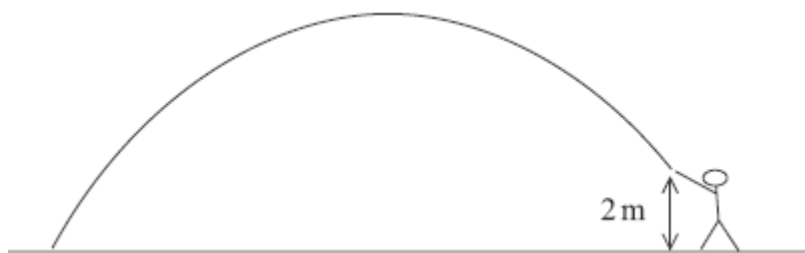
Q18.

A cricket ball is hit at ground level on a horizontal surface. It initially moves at 28 m s^{-1} at an angle of 50° above the horizontal.

- (a) Find the maximum height of the ball during its flight.

(4)

- (b) The ball is caught when it is at a height of 2 metres above ground level, as shown in the diagram.



Show that the time that it takes for the ball to travel from the point where it was hit to the point where it was caught is 4.28 seconds, correct to three significant figures.

(5)

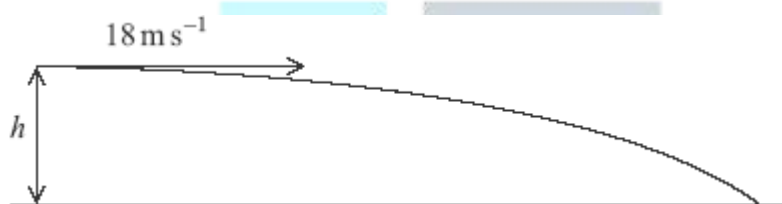
- (c) Find the speed of the ball when it is caught.

(5)

(Total 14 marks)

Q19.

An arrow is fired horizontally at a speed of 18 m s^{-1} from a point at a height of h metres above horizontal ground. The arrow hits the ground after it has been moving for 0.6 seconds. Model the arrow as a particle that moves only under the influence of gravity.



- (a) Show that $h = 1.76$, correct to three significant figures.

(2)

- (b) Find the horizontal distance travelled by the arrow during its flight.

(2)

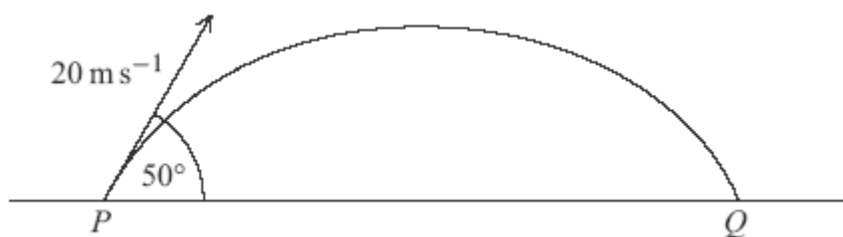
- (c) Find the speed of the arrow and the direction in which it is moving when it hits the ground.

(6)

(Total 10 marks)

Q20.

A ball is kicked from the point P on a horizontal surface. It leaves the surface with a velocity of 20 m s^{-1} at an angle of 50° above the horizontal and hits the surface for the first time at the point Q . Assume that the ball is a particle that moves only under the influence of gravity.



- (a) Show that the time that it takes the ball to travel from P to Q is 3.13 s, correct to three significant figures. (4)
- (b) Find the distance between the points P and Q . (2)
- (c) If a heavier ball were projected from P with the same velocity, how would the distance between P and Q , calculated using the same modelling assumptions, compare with your answer to part (b)? Give a reason for your answer. (2)
- (d) Find the maximum height of the ball above the horizontal surface. (3)
- (e) State the magnitude and direction of the velocity of the ball as it hits the surface. (2)

(Total 13 marks)

Q21.

A particle is projected from a point O on a horizontal plane and has initial velocity components of 2 m s^{-1} and 10 m s^{-1} parallel to and perpendicular to the plane respectively. At time t seconds after projection, the horizontal and upward vertical distances of the particle from the point O are x metres and y metres respectively.

- (a) Show that x and y satisfy the equation Practice. All rights reserved.

$$y = -\frac{g}{8}x^2 + 5x$$

- (b) By using the equation in part (a), find the horizontal distance travelled by the particle whilst it is more than 1 metre above the plane. (4)
- (c) Hence find the time for which the particle is more than 1 metre above the plane. (2)

(Total 10 marks)

Q22.

A pellet is fired from a window at a height of 3 metres above horizontal ground. Initially,

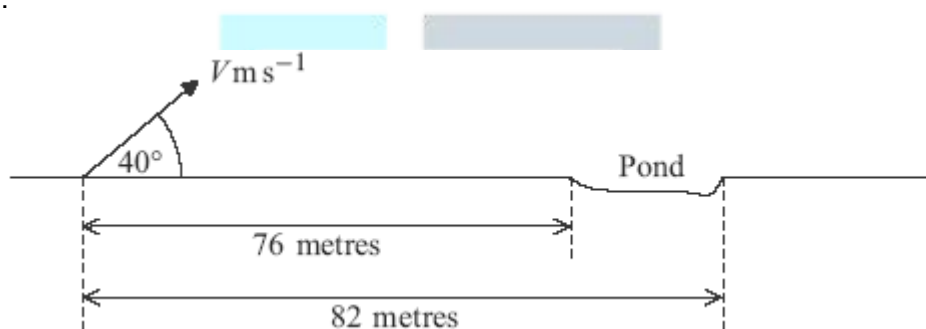
the pellet travels at 70 m s^{-1} at an angle of 10° above the horizontal.

- (a) Show that the time for which the pellet travels before it hits the ground is 2.71 seconds, correct to three significant figures. (6)
- (b) Find the horizontal distance that the pellet travels before it hits the ground. (2)
- (c) Find the minimum speed of the pellet during its flight. (2)
- (d) Find the speed of the pellet when it hits the ground. (4)

(Total 14 marks)

Q23.

A golfer hits a ball which is on horizontal ground. The ball initially moves with speed $V \text{ m s}^{-1}$ at an angle of 40° above the horizontal. There is a pond further along the horizontal ground. The diagram below shows the initial position of the ball and the position of the pond.



- (a) State **two** assumptions that you should make in order to model the motion of the ball. (2)
- (b) Show that the horizontal distance, in metres, travelled by the ball when it returns to ground level is (6)

$$\frac{V^2 \sin 40^\circ \cos 40^\circ}{4.9}$$

- (c) Find the range of values of V for which the ball lands in the pond. (4)

(Total 12 marks)

Q24.

A ball is kicked so that it leaves a horizontal surface, at the point , travelling at 16 m s^{-1} and at an angle θ above the horizontal. The ball hits the surface again 2 seconds

later, at the point B . Assume that the ball is a particle that moves only under the influence of gravity.

- (a) Show that $\theta = 37.8^\circ$, correct to three significant figures.

(3)

- (b) Find the time for which the ball is more than 2 metres above the surface.

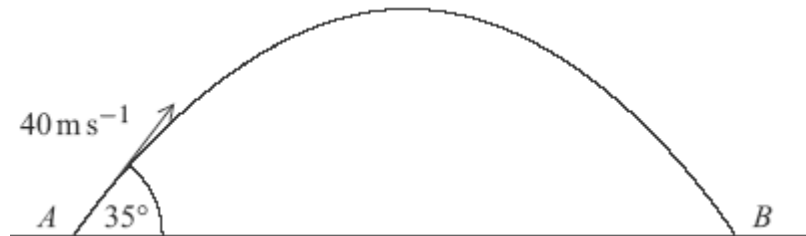
(5)

(Total 8 marks)

Q25.

A ball is hit by a bat so that, when it leaves the bat, its velocity is 40 m s^{-1} at an angle of 35° above the horizontal. Assume that the ball is a particle and that its weight is the only force that acts on the ball after it has left the bat.

- (a) A simple model assumes that the ball is hit from the point A and lands for the first time at the point B , which is at the same level as A , as shown in the diagram.



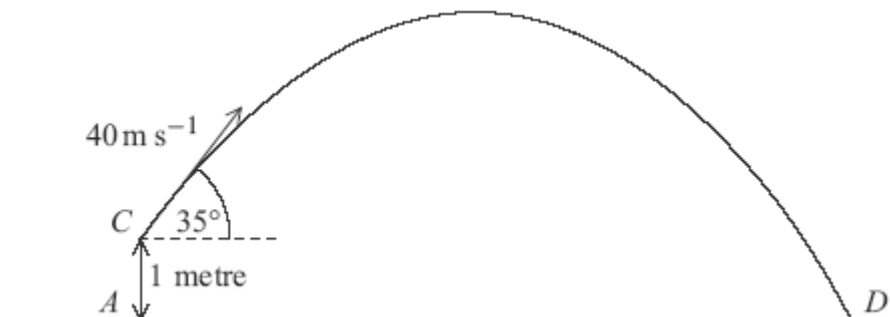
- (i) Show that the time that it takes for the ball to travel from A to B is 4.68 seconds, correct to three significant figures.

(4)

- (ii) Find the horizontal distance from A to B .

(2)

- (b) A revised model assumes that the ball is hit from the point C , which is 1 metre above A . The ball lands at the point D , which is at the same level as A , as shown in the diagram.



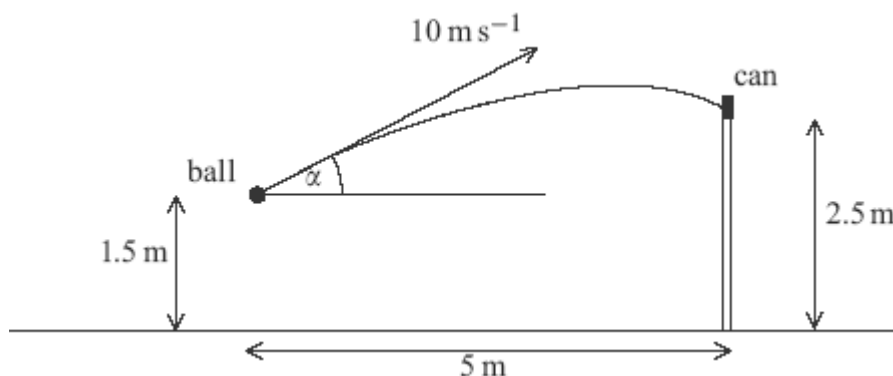
Find the time that it takes for the ball to travel from C to D .

(6)

(Total 12 marks)

Q26.

A boy throws a small ball from a height of 1.5 m above horizontal ground with initial velocity 10 m s^{-1} at an angle α above the horizontal. The ball hits a small can placed on a vertical wall of height 2.5 m, which is at a horizontal distance of 5 m from the initial position of the ball, as shown in the diagram.



- (a) Show that α satisfies the equation

$$49 \tan^2 \alpha - 200 \tan \alpha + 89 = 0$$

(7)

- (b) Find the two possible values of α , giving your answers to the nearest 0.1° .

(3)

- (c) (i) To knock the can off the wall, the horizontal component of the velocity of the ball must be greater than 8 m s^{-1} .

Show that, for one of the possible values of α found in part (b), the can will be knocked off the wall, and for the other, it will **not** be knocked off the wall.

(3)

- (ii) Given that the can is knocked off the wall, find the direction in which the ball is moving as it hits the can.

(4)

(Total 17 marks)

Q27.

A cricket ball is hit from the floor of a sports hall, which has a height of 6 metres. The initial velocity of the ball is 20 m s^{-1} at an angle of 60° above the horizontal.

Assume that the cricket ball is a particle which moves only under the influence of gravity.

- (a) Show that the ball hits the ceiling of the sports hall approximately 0.389 seconds after it was hit.

(5)

- (b) Find the horizontal distance travelled by the ball before it hits the ceiling.

(2)

- (c) Find the speed of the ball just before it hits the ceiling.

(5)

(Total 12 marks)

Q28.

A golf ball is struck from a point on horizontal ground so that it has an initial velocity of 50 m s^{-1} at an angle of 40° above the horizontal.

Assume that the golf ball is a particle and its weight is the only force that acts on it once it is moving.

- (a) Find the maximum height of the golf ball.

(4)

- (b) After it has reached its maximum height, the golf ball descends but hits a tree at a point which is at a height of 6 metres above ground level.



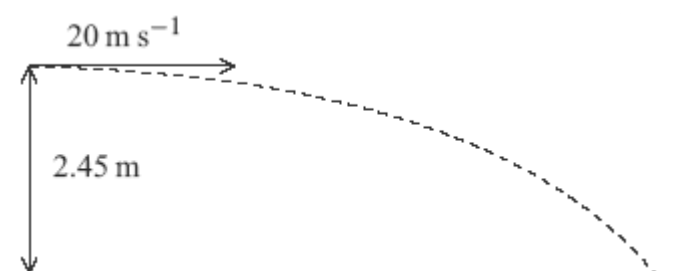
Find the time that it takes for the ball to travel from the point where it was struck to the tree.

(6)

(Total 10 marks)

Q29.

A tennis ball is hit from a height of 2.45 metres above horizontal ground. Initially it travels horizontally at a speed of 20 m s^{-1} , as shown in the diagram.



- (a) Show that the time taken for the tennis ball to reach the ground is 0.707 seconds, correct to three significant figures.

(3)

- (b) Find the horizontal distance travelled by the ball when it hits the ground.

(2)

- (c) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground.

(4)
(Total 9 marks)

Q30.

An arrow is fired from a point A with a velocity of 25 m s^{-1} , at an angle of 40° above the horizontal. The arrow hits a target at the point B which is at the same level as the point A , as shown in the diagram.



- (a) State **two** assumptions that you should make in order to model the motion of the arrow. (2)
- (b) Show that the time that it takes for the arrow to travel from A to B is 3.28 seconds, correct to three significant figures. (4)
- (c) Find the distance between the points A and B . (2)
- (d) State the magnitude and direction of the velocity of the arrow when it hits the target. (2)
- (e) Find the minimum speed of the arrow during its flight. (2)

(Total 12 marks)

Q31.

A ball is projected with speed $u \text{ m s}^{-1}$ at an angle of elevation α above the horizontal so as to hit a point P on a wall. The ball travels in a vertical plane through the point of projection. During the motion, the horizontal and upward vertical displacements of the ball from the point of projection are x metres and y metres respectively.

- (a) Show that, during the flight, the equation of the trajectory of the ball is given by

$$y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha)$$

(6)

- (b) The ball is projected from a point 1 metre vertically below and R metres horizontally from the point P .

- (i) By taking $g = 10 \text{ m s}^{-2}$, show that R satisfies the equation

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0$$

(2)

- (ii) Hence, given that u and R are constants, show that, for $\tan \alpha$ to have real values, R must satisfy the inequality

$$R^2 \leq \frac{u^2(u^2 - 20)}{100}$$

(2)

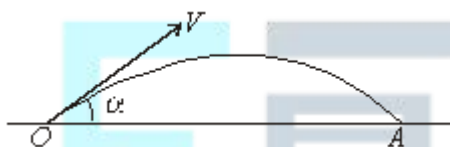
- (iii) Given that $R = 5$, determine the minimum possible speed of projection.

(3)

(Total 13 marks)

Q32.

A golf ball is projected from a point O with initial velocity V at an angle α to the horizontal. The ball first hits the ground at a point which is at the same horizontal level as O , as shown in the diagram.



It is given that $V \cos \alpha = 6u$ and $V \sin \alpha = 2.5u$.

- (a) Show that the time taken for the ball to travel from O to A is $\frac{5u}{g}$.

(4)

- (b) Find, in terms of g and u , the distance OA .

(2)

- (c) Find V in terms of u .

(2)

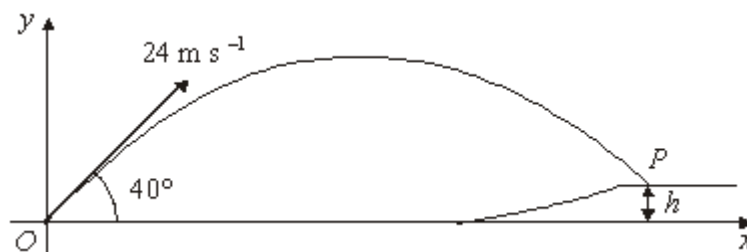
- (d) State, in terms of u , the least speed of the ball during its flight from O to A .

(1)

(Total 9 marks)

Q33.

A golf ball is struck from a point O with velocity 24 m s^{-1} at an angle of 40° to the horizontal. The ball first hits the ground at a point P , which is at a height h metres above the level of O .



The horizontal distance between O and P is 57 metres.

- (a) Show that the time that the ball takes to travel from O to P is 3.10 seconds, correct to three significant figures.

(3)

- (b) Find the value of h .

(3)

- (c) (i) Find the speed with which the ball hits the ground at P .

(5)

- (ii) Find the angle between the direction of motion and the horizontal as the ball hits the ground at P .

(2)

(Total 13 marks)

Q34.

A football is kicked from a point O on a horizontal football ground with a velocity of 20 m s^{-1} at an angle of elevation of 30° . During the motion, the horizontal and upward vertical displacements of the football from O are x metres and y metres respectively.

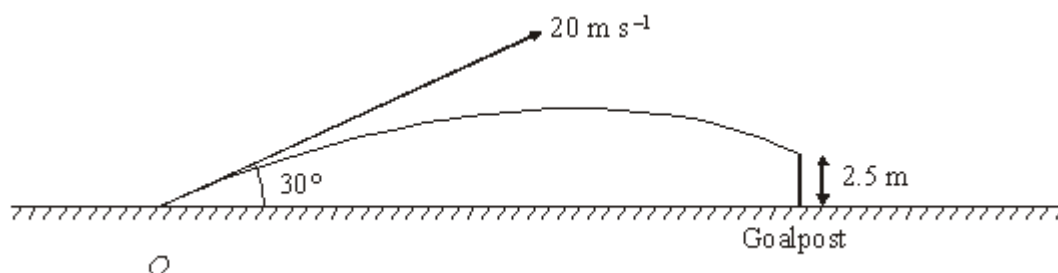
- (a) Show that x and y satisfy the equation

$$y = x \tan 30^\circ - \frac{gx^2}{800 \cos^2 30^\circ}$$

(6)

- (b) On its downward flight the ball hits the horizontal crossbar of the goal at a point which is 2.5 m above the ground. Using the equation given in part (a), find the horizontal distance from O to the goal.

(4)



- (c) State **two** modelling assumptions that you have made.

(2)

(Total 12 marks)

Q35.

A golf ball is struck so that its initial velocity is 30 m s^{-1} at an angle of 60° above the horizontal. Assume that the ground is horizontal.

- (a) (i) Show that the time of flight of the particle is 5.30 seconds, correct to three significant figures.

(3)

- (ii) Find the distance of the ball from its initial position when it hits the ground for the first time.

(2)

- (b) In fact, the ball hits a tree at a point which is at a height of 5 metres above the ground. Given that the ball is descending when it hits the tree, calculate the distance of the tree from the point where the ball was struck.

(6)

(Total 11 marks)



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